

Wheeled Locomotion

Autonomous Navigation of Mobile Robots

Anna Semakova

a.semakova@gmail.com

20.02.2025

Locomotion

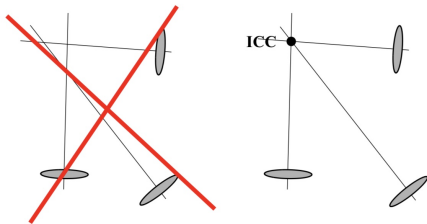
of Wheeled Robots

Locomotion (Oxford Dictionary) – movement or the ability to move

- Differential drive (AmigoBot, Pioneer 2-DX)
- Car drive (Ackerman steering)
- Synchronous drive (B21)
- Mecanum wheels

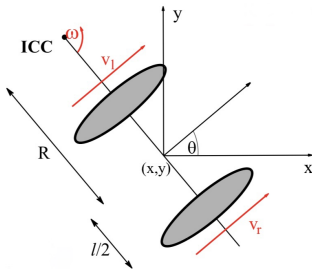
ICC

Instantaneous Center of Curvature



- For rolling motion to occur, each wheel has to move along its y -axis

Differential Drive



$$\text{ICC} = [x - R \sin \theta; y + R \cos \theta]$$

$$v_r = \omega (R + l/2)$$

$$v_l = \omega (R - l/2)$$

$$R = \frac{l}{2} \cdot \frac{v_r + v_l}{v_r - v_l}$$

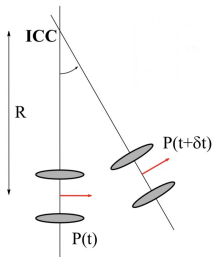
$$\omega = \frac{v_r - v_l}{l}$$

$$v = \frac{v_r + v_l}{2}$$

Differential Drive

Forward Kinematics

$$\begin{bmatrix} x' \\ y' \\ \theta' \end{bmatrix} = \begin{bmatrix} \cos(\omega\delta t) & -\sin(\omega\delta t) & 0 \\ \sin(\omega\delta t) & \cos(\omega\delta t) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x - \text{ICC}_x \\ y - \text{ICC}_y \\ \theta \end{bmatrix} + \begin{bmatrix} \text{ICC}_x \\ \text{ICC}_y \\ \omega\delta t \end{bmatrix}$$



$$x(t) = \int_0^t v(t') \cos(\theta(t')) dt'$$

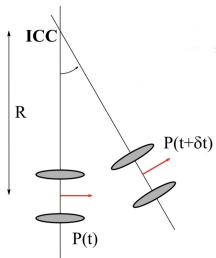
$$y(t) = \int_0^t v(t') \sin(\theta(t')) dt'$$

$$\theta(t) = \int_0^t \omega(t') dt'$$

Differential Drive

Forward Kinematics

$$\begin{bmatrix} x' \\ y' \\ \theta' \end{bmatrix} = \begin{bmatrix} \cos(\omega \delta t) & -\sin(\omega \delta t) & 0 \\ \sin(\omega \delta t) & \cos(\omega \delta t) & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} x - \text{ICC}_x \\ y - \text{ICC}_y \\ \theta \end{bmatrix} + \begin{bmatrix} \text{ICC}_x \\ \text{ICC}_y \\ \omega \delta t \end{bmatrix}$$

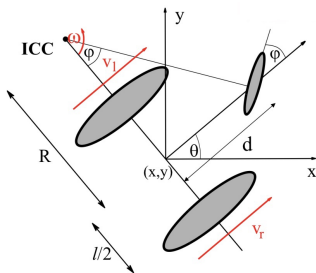


$$x(t) = \frac{1}{2} \int_0^t (v_r(t') + v_l(t')) \cos(\theta(t')) dt'$$

$$y(t) = \frac{1}{2} \int_0^t (v_r(t') + v_l(t')) \sin(\theta(t')) dt'$$

$$\theta(t) = \frac{1}{l} \int_0^t (v_r(t') - v_l(t')) dt'$$

Ackermann Drive



$$\text{ICC} = [x - R \sin \theta; y + R \cos \theta]$$

$$R = \frac{d}{\tan \phi}$$

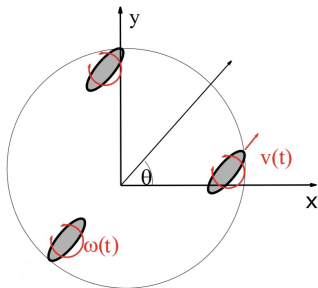
$$v_r = \omega (R + l/2)$$

$$v_l = \omega (R - l/2)$$

$$R = \frac{l}{2} \cdot \frac{v_r + v_l}{v_r - v_l}$$

$$\omega = \frac{v_r - v_l}{2}$$

Synchronous Drive



$$x(t) = \int_0^t v(t') \cos(\theta(t')) dt'$$

$$y(t) = \int_0^t v(t') \sin(\theta(t')) dt'$$

$$\theta(t) = \int_0^t \omega(t') dt'$$

Mecanum Wheels



$$v_y = (v_0 + v_1 + v_2 + v_3) / 4$$

$$v_x = (v_0 - v_1 + v_2 - v_3) / 4$$

$$v_\theta = (v_0 + v_1 - v_2 - v_3) / 4$$

$$v_{error} = (v_0 - v_1 - v_2 + v_3) / 4$$

Non-Holonomic Constraints

- Non-holonomic constraints limit the possible incremental movements within the configuration space of the robot
- Robots with differential drive or synchro-drive move on a circular trajectory and cannot move sideways
- Mecanum-wheeled robots can move sideways (they have no non-holonomic constraints)

Holonomic vs. Non-Holonomic

- Non-holonomic constraints reduce the control space with respect to the current configuration
 - E.g., moving sideways is impossible
- Holonomic constraints reduce the configuration space
 - E.g., a train on tracks (not all positions and orientations are possible)

Dead Reckoning and Odometry

- Estimating the motion based on the issued controls/wheel encoder readings
- Integrated over time

